

Saiprakash Nalubolu

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Summary

ISTQB certified SQA Engineer with 4 years across Fintech and Healthcare, specializing in UI and API automation testing. Has published AI research, hands-on product development building agentic AI systems, owning QA lifecycle across web and mobile.

Technical skills

Languages: Python, Java, SQL, Git, Linux, Gherkin

Platforms & Delivery: Selenium, Playwright, Cucumber/Behave(BDD), Bruno, Postman, SoapUI, PCOMM, HP ALM, Xray, Docker, REST APIs, SQLdeveloper, PostgreSQL, Faktory; CI/CD (Jenkins, GitHub Actions) Maven, AWS, ARD(Agile Requirement Designer)

AI/ML & LLM: PyTorch, TensorFlow/Keras, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Transformers(BERT, RoBERTa), LangChain, RAG, embeddings/vector search, CUDA/cuDNN, OpenCV, SHAP, CNN/CRNN, PyTorch Mobile

Experience

Software QA Engineer, MiHIN

Sep 2025 – Present

- Built Selenium + Behave(BDD) automation from scratch for Diretto, a healthcare website supporting interoperability across Michigan's Health Information Network (MiHIN).
- Developed a Playwright automation suite to validate FHIR APIs and HL7 v2 message logs after dispatch through the HIE router
- Reduced manual testing effort by 45% and improved QA sign off timeline by 66% by building scalable UI automation workflows
- Monitored and triaged interoperability defects using AWS CloudWatch logs and Grafana dashboards for root cause analysis.

Research Assistant, Binghamton University

Sep 2024 - Feb 2025

- Published [ClinSegAI](#) a post processing framework for Histopathology Image segmentation achieving 0.80 DSC and best HD95/ASSD vs 6 leading tools. Automated WSI segmentation workflow, cutting image preprocessing time by 33%.
- Built a lung cancer survival model by fusing TCGA clinical + histopathology features, explained with SHAP (TNM, ECOG, NLR/PLR, morphology), and scaled training on HPC GPUs using PyTorch FSDP + CUDA (A100/H100/T4).

QA Automation Engineer, Citibank (Through TCS)

Mar 2021 – Dec 2023

- Led Functional, Integration, End-to-End and Regression testing for Citibank customer service website across 17 different projects.
- Increased automation test coverage from 66% to 87% in 2 years by architecting modular, reusable automation test suites
- Developed Cucumber BDD frameworks with Page Object Model (POM), boosting reusability and reducing maintenance by 24%.
- Designed and executed 1,000+ test cases, uncovering 400+ defects thus improving release stability.
- Integrated automation into Jenkins CI/CD, enabling daily automated regressions and faster build verification.
- Collaborated with developers, product owners and Business Analysts thus accelerating critical defect resolution.
- Led a team of 5 to ensure accurate communication to 30M+ users by automating customer servicing message workflows.

Education

Binghamton University, State University of New York

Jan 2024 - Dec 2025

Master of Science in Computer Science(AI track)

CGPA: 3.84/4.00

Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Human Computer interaction, Design patterns.

Vellore Institute of Technology (VIT), Vellore

Jul 2016 - Jun 2020

Bachelor of Technology in Computer Science and Engineering

Projects

Baaraakatta: Board game | [Playstore link](#)

- Leveraging Claude Code, I shipped the game to Google Play; owned QA lifecycle for real time multiplayer game (Web and Mobile)
- Validated game behavior across web and Android for responsive layouts, native OAuth, and platform-specific edge cases

Kitchbot: Agentic Meal Planner

- Architected a ReAct agent (LangChain + Qwen) with Pydantic-enforced tool contracts for reliable LLM routing.
- Validated agentic workflows across LLM tool calls, Supabase auth, and Spoonacular API integration end-to-end.
- Reduced token costs via conversation summarization; enforced rate limits and validated API caching in production.

Samethalu.com: RAG search for Telugu Proverbs

- Live RAG service handling 100-250 daily queries; maps free text situations to proverbs via a low latency API + web UI.
- Built embeddings + vector search with metadata filters(translation, meaning, usage) for precise, explainable top k matches.
- Docker containerized API + web UI with usage analytics and multilingual search for relevance tuning.

Achievements

- Top [1% Global finish in TCS codeVita S13\(2025-26\)](#) a global coding contest recognized by Guinness book of world records.
- Received Best employee of the month award 3 times at Tata Consultancy Services Ltd (CITI Bank QA division).
- Published a [peer reviewed article](#) in Computers in Biology and Medicine by Elsevier(ScienceDirect) vol 199 on cell segmentation.